Let the horizontal plane containing the fixed point P be the zero potential energy reference level (with the downward direction as the positive y-axis). The rope is uniform with a linear mass density of $\rho = \frac{2M}{2L} = \frac{M}{L}$.

1. Initial State ($t = 0$):

- Initial kinetic energy of the system: $E_{k0} = 0$
- Initial potential energy of the system (center of mass of the rope falls by $\frac{L}{2}$): $E_{p0} = -2Mg\left(\frac{L}{2}\right) = -MgL$
- **Total initial mechanical energy:** $E_0 = -MgL$

2. Final State (particle has fallen a distance x):

- **Final kinetic energy:** The particle and the moving section of the rope have the same speed v.

$$E_k = \frac{1}{2}mv^2 + \frac{1}{2}\left[\frac{M}{L}\left(L - \frac{x}{2}\right)\right]v^2 = \frac{1}{2}\left(m + M - \frac{Mx}{2L}\right)v^2$$

- **Final potential energy:** Composed of three parts: the particle, the stationary section of the rope (length $L + \frac{x}{2}$), and the moving section of the rope (length $L - \frac{x}{2}$).

$$E_p = -mgx - \underbrace{\frac{Mg}{L}\left(L + \frac{x}{2}\right)}_{\text{mass of stationary section}} \underbrace{\frac{1}{2}\left(L + \frac{x}{2}\right)}_{\text{COM of stationary section}} - \underbrace{\frac{Mg}{L}\left(L - \frac{x}{2}\right)}_{\text{mass of moving section}} \underbrace{\left(\frac{L}{2} + \frac{3x}{4}\right)}_{\text{COM of moving section}}$$

Expanding and simplifying yields:

$$E_p = -mgx - MgL - Mgx + \frac{Mgx^2}{4L}$$

3. Conservation of Mechanical Energy ($E_0 = E_k + E_p$):

$$-MgL = \frac{1}{2}\left(m + M - \frac{Mx}{2L}\right)v^2 - mgx - MgL - Mgx + \frac{Mgx^2}{4L}$$

$$\frac{1}{2}\left(\frac{mL + ML - \frac{1}{2}Mx}{L}\right)v^2 = gx(m + M) - \frac{Mgx^2}{4L}$$

Extracting the common factor and rearranging proves the expression:

$$v^2 = \frac{2gx(mL + ML - \frac{1}{4}Mx)}{mL + ML - \frac{1}{2}Mx}$$

Using the kinematic differential relationship $a = v\frac{dv}{dx} = \frac{1}{2}\frac{d(v^2)}{dx}$, we differentiate the above expression for v^2.

Let $v^2 = \frac{u}{w}$, where:

- Numerator $u = 2gL(m + M)x - \frac{1}{2}Mgx^2 \implies u' = 2gL(m + M) - Mgx = 2gw$

- Denominator $w = L(m+M) - \frac{1}{2}Mx \implies w' = -\frac{1}{2}M$

According to the quotient rule:

$$\frac{d(v^2)}{dx} = \frac{u'w - uw'}{w^2} = \frac{(2gw)w - u(-\frac{1}{2}M)}{w^2} = 2g + \frac{Mu}{2w^2}$$

Substituting into $a = \frac{1}{2}\frac{d(v^2)}{dx}$:

$$a = g + \frac{Mu}{4w^2}$$

Substituting the original expressions for u and w back and simplifying proves the expression:

$$a = g + \frac{Mgx(mL + ML - \frac{1}{4}Mx)}{2\left(mL + ML - \frac{1}{2}Mx\right)^2}$$

Observe the expression for acceleration:

$$a = g + \frac{Mgx(mL + ML - \frac{1}{4}Mx)}{2\left(mL + ML - \frac{1}{2}Mx\right)^2}$$

At any instant after the particle is released ($x > 0$):

1. **Denominator:** $2\left(mL + ML - \frac{1}{2}Mx\right)^2$ is a perfect square and is always strictly positive (> 0).
2. **Numerator:** The problem states that $x < 2L$, therefore $\frac{1}{4}Mx < \frac{1}{4}M(2L) = \frac{1}{2}ML$. From this, we can deduce the term inside the parenthesis:

$$(mL + ML - \frac{1}{4}Mx) > \left(mL + ML - \frac{1}{2}ML\right) = mL + \frac{1}{2}ML > 0$$

Since M, g, x are all positive numbers, and the term inside the parenthesis is also positive, the entire fractional term is strictly greater than 0. Therefore:

$$a = g + (\text{positive number}) > g$$

The original proposition is proven.

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